

The empirical recurrence for OEIS A234477: recovering a conjecture that is not written down

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Abstract

OEIS A234477 records “Empirical recurrence of order 46”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 512$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 46, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 46 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234477 is “Number of $(n+1) \times (2+1)$ 0..7 arrays with every 2×2 subblock having its diagonal sum differing from its antidiagonal sum by 1 (constant-stress 1×1 tilings)”. It has offset 1 and begins

7840, 141728, 2765648, 56172656, 1165600828, 24484032192, 517788509004, 10996717871928, ...

The entry states:

Empirical recurrence of order 46 (see link above).

As of the “Last modified” line on the live entry (Jun 20 2022 01:26:01, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 512$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 512$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1024$ exact terms of the model returns order 46 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{46} c_i a(n-i),$$

c_1	44	c_{13}	−1422314410108894	c_{25}	−363883699796527015801344	c_{37}	50471854
c_2	457	c_{14}	−29152136220988528	c_{26}	596869370049421248325632	c_{38}	69437861
c_3	−44620	c_{15}	128401145159161040	c_{27}	1998274253544647331375104	c_{39}	−4286165
c_4	234930	c_{16}	962583292724269872	c_{28}	−4470874121743656543588352	c_{40}	−2003216
c_5	16248415	c_{17}	−6042814510053844248	c_{29}	−6787171179199293038534656	c_{41}	16590217
c_6	−185703379	c_{18}	−19260404475359093056	c_{30}	20473090752611461659754496	c_{42}	13742352
c_7	−2778832058	c_{19}	176642861570874277792	c_{31}	12492902141433622283501568	c_{43}	−2408545
c_8	48457533125	c_{20}	189199164561946063232	c_{32}	−58778376409754117102141440	c_{44}	1377813
c_9	213875519821	c_{21}	−3380781508738075730624	c_{33}	−5498128373084365375471616	c_{45}	1073431
c_{10}	−6716954885396	c_{22}	588511898385060076032	c_{34}	105166618594294247765311488	c_{46}	−109854
c_{11}	130034032028	c_{23}	43060005266709461147392	c_{35}	−23797589538129121646739456		
c_{12}	558392797091904	c_{24}	−43268532698755630371328	c_{36}	−113836957249785716910063616		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 46, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $46 + 4$ terms removed returns order 46 as well, so $\deg q_{\min} = 46 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $46 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 46 that this sequence satisfies — and that family turns out to have exactly one member.

References

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